GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	**Pipeline Pre-commissioning Specification**
Document Number	EA-BFPCL-GPMC-GGPL-FEED-PSE-PLN-003
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Pipeline Pre-commissioning Specification

R01	01.06.22	Issued for Review	B. Akinwale	Kazeem Yusuff	Aloy Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	01.06.22	First Issue for Clients Review
R02		
A01		

Review Code & Status (to be filled by Client)		
☐ Code 1	**Approved** – Further work may proceed.	
☐ Code 2	**Approved with comments.** – Incorporate Comments. Up-Rev & Resubmit. Work May Proceed	
☐ Code 3	**Reviewed** – Incorporate Comments. Up-Rev & Resubmit. Work May Not Proceed	
☐ Code 4	**Retained** for information only	
☐ Code 5	**Review Not Required** – Do not Resubmit	
Date:	Position:	Sign:

Revision Philosophy:

a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	**Project No.** 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	**Doc No.** EA-BFPCL-GPMC-GGPL-FEED-PSE-PLN-003

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES	
Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA/SPDC-BFPC UGS-GEP/CSM-	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-	Conceptual Study Memorandum
EA/SPDC-BFPC UGS-GEP/SVY-RPT	Pipeline Route Selection Study
EA/SPDC-BFPC UGS-GEP/PRC-	Process Simulation Study – HYSYS, PIPESIM, & OLGA
EA/SPDC-BFPC UGS-GEP/MAT-	Pipeline Materials Selection & Corrosion Analysis Study
EA/SPDC-BFPC UGS-GEP/ELE-RPT-	Manifold Platforms Power Supply Options
EA/SPDC-BFPC UGS-GEP/MEC-	Manifold Platforms Concept & Integration with SPDC
EA/SPDC-BFPC UGS-GEP/PRC-	Gas Processing Technology Option
6064/PLRS/001a-PRP	Onshore Pipeline Route Survey for BFCL; Topographical Surveys
40km Gas Supply Pipeline Project ESIA	Environmental and Social Impact Assessment Report prepared by Messrs. Fundamental Integrated Site Appraisal Services Limited - Draft Report November 2020.
DEP 31.40.40.38-Gen.	Hydrostatic pressure testing of new pipelines
DEP 33.64.10.10-Gen.	Electrical engineering guidelines
DEP 61.40.20.30-Gen	Welding of pipelines and related facilities

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Abbreviations

Abbreviation	Meaning
AASHTO	American Association of state Highway Transportation Officials
AGEF	Above Ground End Facilities
ALARP	As Low as Reasonably Practicable
API	American Petroleum Institute
ASTM	American Society of Testing and Materials
BOD	Basis Of Design
CP	Cathodic Protection
CPT	Cone Penetrometer Test
CTMS	Custody Transfer Metering Station
DB&B	Double Block and Bleed
DCQ	Daily Contractual Quantity
DEP	Design and Engineering Practice
ESD	Emergency Shutdown
ESDV	Emergency Shutdown Valve
FEED	Front End Engineering Design
FGS	Fire, Gas and Smoke detection systems
GPP	Gas Processing Plant
GTG	Gas Turbine Generators
HAZID	Hazard Identification
HAZOP	Hazard Operability
HEMP	Hazards and Effects Management Process
HFE	Human Factors Engineering
HPU	Hydraulic Power Unit
HRA	Health Risk Assessment
HSES	Health, Safety, Environment and Security
HSSE & SP	Health, Safety, Security, Environment and Social Performance
IEC	International Electro-technical Commission
ISPM	Industry Standard Process Modules
JV	Joint-Venture

LCD	Liquid Crystal Display
LOI	Local Operator Interface
LRL	Lower Range Limit
LRV	Lower Range Value
LIRA	Logistics and Infrastructure Resource Assessment
MAC	Main Automation Contractor
NAG	Non-Associated Gas
OPC	OLE for Process Control
OPPS	Overpressure Protection System
PCD	Process Control Domain
PDMS	Plant Design and Management System
PEPF	Package Engineering, Procurement, and Fabrication
PIG	Pipeline Inspection Gauge
PVC	Polyvinyl Chloride
RTR	Reliability Test Run
SCE	Safety critical elements
SI	International System Units
SIL	Safety Integrity Level
SRB	Sulphate Reducing Bacteria
SIS	Safety Instrumented System
SWB	Steel Wire Braiding
URV	Upper Range Value
USC	US Customary Units
XLPE	Cross Linked Polyethylene
TEG	Tri-ethylene Glycol
TQ	Top Quartile

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

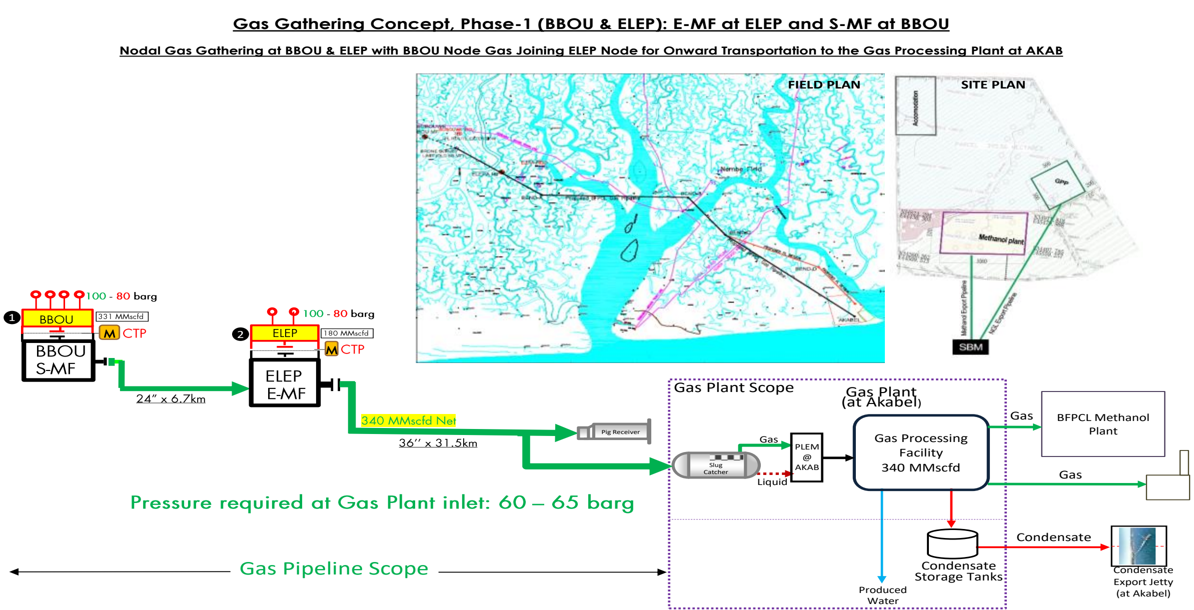

Figure 1: Gas Gathering Concept, Phase 1

2. Regulations, Codes And Standards

2.1. Regulations

The following listed regulations in the below table will be used in the project as applicable.

Regulation	Function
Petroleum Act, CAP P10, Laws of the Federation of Nigeria, LFN 2004, amended.	This is BFPCL statute regulating the oil industry. The following regulation relevant to this project is issued pursuant to the Act.
Mineral Oil (Safety) Regulations 1963, as amended.	These regulations, in so far as they apply to pipelines, require compli-ance with relevant IP, API and ASME codes and standards.
Oil Pipelines Act, 1956. CAP. 145 (Amended 1965, CAP. 338, LFN 1990)	Provides for licenses to be granted for the establishment and mainte-nance of pipelines incidental and supplementary to oilfields and oil mining, and for purposes ancillary to such pipelines.
Oil and Gas Pipelines Regulations, 1995	Details the regulations for the design, construction and maintenance of oil and gas pipelines.
Environmental Impact Assessment Act, CAP E12, LFN 2004.	Makes the conduct of an EIA mandatory prior to the development of any project or activity likely to have a significant effect on the envi-ronment
Federal Environment Protection Agency Act, CAP F13, LFN 2004.	Provides for the protection of the environment and sustainable devel-opment of Nigeria's natural re- sources.
Environmental Guidelines and Standards For The Petroleum Indus try In Nigeria, 1991.	DPR (NUPRC) issued document providing guidelines on pollution abatement and relevant legislation.

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Pipeline Pre-commissioning Specification

3.1. General Requirements

This document provides the step-by-step requirements preceding commissioning of the natural gas pipelines. The contractor shall perform the pre-commissioning of the pipelines in accordance with the requirements of this specification and as contained in the Shell DEP on Pre-commissioning of Pipelines, and or as may be otherwise provided by COMPANY through its representative.

3.1.1. Requirements for Pre-commissioning

Prior to commencing the pre-commissioning activities, the following requirements as a minimum must be satisfied and adhered to:

- The Contractor shall perform the pre-commissioning in accordance with the requirements of the scope of work and the referenced DEP. The scope of work shall take precedence in case of conflict between the requirements of this Specification and the scope of work. Pre-commissioning should commence either immediately after completion of the hydrostatic pressure test or immediately prior to commissioning as specified in the scope of work.
- Pre-commissioning activities shall be executed in the sequence defined in the flowchart in the referenced DEP unless otherwise specified in the scope of work.
- The following general information shall be provided in the scope of work for pre-commissioning:

    - all relevant pipeline data;
    - hydrotest water quality;
    - the existing hazardous area classifications of the proposed work sites.

- Hydrotest water and all water required for the pre-commissioning shall be supplied and disposed of in accordance with DEP 31.40.40.38-Gen.
- NOTE: DEP 31.40.40.38-Gen. also specifies the requirements for water filtration and treatment.

    - Media other than water that will be introduced into the pipeline for pre-commissioning shall meet the following requirements:
    - they shall be clean and free of oil;
    - they shall not have any detrimental impact on the pipeline materials, including internal coating, if applicable;
    - nitrogen, methanol and glycol shall have a purity of at least 99 %;
    - the temperature of the media when entering the pipeline shall not be less than 0oC or the minimum pipeline design temperature, whichever is the highest, and not more than

50oC or the maximum pipeline design temperature, whichever is the lowest.

- All measurements, calculations and reporting shall be in SI units. Pressures shall be quoted in bar (g) except for vacuum pressures which shall be stated in bar (abs).

3.1.2. Pre-commissioning Plan

- The Contractor shall execute the work in accordance with the pre-commissioning plan approved by BFPCL before commencement of the work.
  The Contractor shall prepare the final pre-commissioning plan for submission to BFPCL for approval not later than 4 weeks or as may be agreed before commencement of the work unless otherwise specified in the scope of work. An outline or draft pre-commissioning plan should have been agreed as part of the contract award process.
- The pre-commissioning plan shall include as a minimum:

  - an outline description of the required pre-commissioning work including sequence of activities and programme. For the individual activities, the programme should include the start and completion dates, and the dates of mobilisation and demobilisation of equipment, personnel and supplies to the work sites;
  - the Contractor's organisation chart for the work with a description of the responsibilities of key personnel;
  - names of any subcontractors and Suppliers and their scope of work or supply, and how they shall be managed by the Contractor.
  - description of the interfaces and lines of communication with BFPCL.
  - identification of all potential health, safety, and environmental hazards with a description of the contingency procedures and an emergency plan
  - a procedure for each main activity covering:

i. a step-by-step description of the work.
ii. description of the required personnel, equipment and consumables, certification, monitoring, records, and verifications.
iii. location and layouts with hazardous area classifications for equipment and supplies, including water supply and disposal facilities.
iv. calculations and any other evidence necessary to demonstrate that the work will be executed in accordance with the requirements of BFPCL.

  - contingency procedure for the removal of stuck pigs if used.
  - the "Table of Contents" for the "Final Pre-commissioning Report" with sample copies of Contractor's standard forms and sheets.

3.1.3. Checks and Compliance

- The Contractor shall verify throughout that all the work is executed in full compliance with all the requirements of the scope of work and this Specification. Verifications shall, as a minimum, include:

    - availability of the approved pre-commissioning plan at the work site before commencement of and during the work;
    - identification, certification and condition of equipment and supplies to be provided in good serviceable condition, at the work site prior to mobilisation and throughout pre-commissioning;
    - positioning of necessary warning notices, marker tapes, protective barriers, safety and firefighting equipment;
    - compliance with the hazardous area requirements;
    - installation, earthing and protection of equipment;
    - compliance with safe storage requirements of chemicals;
    - instruction of personnel on their responsibilities and tasks and possible hazards with the necessary safety measures;
    - availability and functioning of safety and communication equipment, including backup equipment;
    - availability of required permits and adherence to their requirements.

3.1.4. Pre-commissioning Records and Reports

- The Contractor shall take the records specified in the scope of work and this Specification and keep them for inclusion in the "Final Pre-commissioning Report". All record charts shall be signed by the Contractor immediately before they are placed in and after they are taken out of the recorder.
- The Contractor shall prepare a "Final Pre-commissioning Report" which shall include as a minimum:

    - overall debrief of the pre-commissioning work;
    - analysis of performance against the pre-commissioning plan;
    - records specified in this DEP.

- The draft of the report shall be submitted for approval to BFPCL within 14 days following the completion of the site work. In the event of comments by BFPCL, the Contractor shall update the report within 14 days.
- The Contractor shall issue one final report with all original records/documents and three

copies (or more if specified in the scope of work) within 14 days of approval of the draft report by BFPCL.

4. Health, Safety and Environment

The Contractor shall, in the pre-commissioning plan, identify the hazards associated with the pre-commissioning activities and work sites, including those advised by BFPCL, and shall define the necessary hazard mitigation measures. The Contractor shall comply with any permit-to-work system in force at any location affected by the pre-commissioning activities.

The Contractor shall provide gas detection devices and/or breathing apparatus when required for safe working practices and when specified in the scope of work. Users of such equipment shall be thoroughly trained in their use.

Equipment should be installed at locations away from public places, roads or living quarters. Contractor shall control access to and activities at these locations.

The Contractor shall place warning signs and notices, such as 'KEEP AWAY - PIPELINE UNDER PRESSURE' or 'KEEP AWAY - PIPELINE UNDER VACUUM', at appropriate places such as affected pipework, isolation valves in branches and main line block valve stations. Notices shall be in English and the local or working languages.

The Contractor shall provide radio or other means for dedicated communication between locations where pre-commissioning work is being carried out or that are affected by Shell work.

The Contractor shall ensure that the hazards associated with the use of nitrogen and the safety precautions to protect personnel are known, understood and applied by all personnel carrying out the work.

Material safety data sheets shall be available at the site where chemicals are stored or handled. Chemicals shall be stored, handled, used, recycled and disposed of in accordance with the recommendations of the Supplier and other applicable requirements. Chemicals should be stored in properly marked original packings, in small quantities and away from sources of heat and fire. Contractor shall inform all personnel at the work sites of the hazards and provide the necessary training and protective means. The Contractor shall be responsible for carrying out any Chemical Hazard Assessment and Risk Assessment required by local authorities before using the proposed chemicals.

Soil and water contamination from spilled liquids shall be prevented. Drip pans shall be placed underneath pig trap doors, engine driven equipment, drains and other places of possible liquid spillage to collect liquids which may cause contamination.

The unsafe release of toxic, flammable and explosive gas mixtures shall be prevented by providing facilities for safe venting or disposal as appropriate.

Noise emission levels shall be kept as low as practical by the selection of appropriate equipment and by attenuation measures such as noise absorbing barriers and silencers. Maximum permissible levels for safe working, as specified in the scope of work, shall not be exceeded. The Contractor shall at all times ensure that the noise level does not exceed 85 dB in areas accessible to the general public unless otherwise agreed by BFPCL.

The Contractor shall be responsible for obtaining all necessary approvals and permits from the appropriate local authorities, owners and third parties concerning the disposal of water from the pipeline. Water may only be directly discharged into the environment after it has been filtered/separated and neutralised, and the environmental acceptability of the effluent has been confirmed.

The location of the pre-commissioning work sites and the arrangement of the equipment (including water supply and disposal sites and facilities) shall be approved by BFPCL.

The Contractor shall prepare an emergency plan which defines the actions, personnel and facilities required in case of an emergency. The Contractor shall obtain BFPCL's approval in writing of the emergency plan before commencing pre-commissioning activities. The plan shall address at least the following aspects:

- handling of spillages and other incidents;
- requirements for personnel and equipment;
- requirements for informing others;
- first aid and medical facilities;
- telephone numbers of stand-by crews;
- telephone numbers of police, medical and other authorities to be contacted;
- telephone number(s) of the contact(s) of BFPCL.

The pre-commissioning work sites shall be left in a clean condition and free of debris. All waste, chemicals, etc. shall be disposed of by the Contractor in accordance with the principal's waste disposal requirements.

5. Precommissioning Equipment

5.1. Pressure or Vacuum Retaining Equipment

All pressure or vacuum retaining pre-commissioning equipment shall be designed, fabricated, tested and marked, in accordance with recognised standards, for a safe operating pressure of not less than the maximum/minimum pressure predicted during the pre-commissioning. Certificates, approved by an independent certifying authority, specifying the applicable standards, confirming compliance with them and stating the hydrostatic pressure test data, shall be provided with all such equipment. The certificates shall be available at the work sites at all times.

5.2. Filters, Pumps and Compressors

The Contractor shall define the number, size and performance of each component required for the work, including standby units and spare parts. Standby compressors should be provided because compression may not be shut down during dewatering due to the risk of breaking the water column and the resulting formation of accumulated heads. Pumps and compressors shall be supplied with test performance data sheets.

Delivery pressures of pumps and compressors shall not at any time exceed the MAOP of the pipeline and of the pre-commissioning equipment, including allowances required for transient pressures e.g. due to surges and static head effects. Pumps and compressors shall be fitted with dedicated over-pressure protection devices if the maximum delivery pressure they can generate exceeds the MAOP. 50 μm water filters, or a size specified in the scope of work, with a capacity sufficient for the required flowrates shall be provided. It should be possible to clean the filters without interrupting operation.

All equipment shall be suitable for continuous use at the required operating conditions under the specified local environmental conditions.

5.3. Pig Traps

Where available and agreed by BFPCL, permanent pig trap facilities may be used for pre-commissioning. If pre-commissioning is carried out in the reverse direction to normal operations, the Contractor shall be responsible for identifying and providing any temporary modifications and/or additional facilities that may be required.

Any temporary pig trap facilities required shall be designed, fabricated, installed and tested in accordance with the same code as the pipeline and shall be suitable for the worst conditions that may arise during pre-commissioning. Temporary pig traps shall be provided with a major barrel capable of storing at least three pigs, unless otherwise specified in the scope of work. They shall be connected to the pipeline by flanging to permanent pipeline flanges, or by welding.

5.4. Welding of Temporary Equipment

Requirements for the welding of temporary pig trap facilities to the permanent pipeline are:

- connecting welds that have not been subjected to the hydrostatic pressure test of the pipeline shall be treated as "golden welds". Such welds go through extensive non-destructive testing (NDT) to ensure they are defect-free in line with standards.
- the minimum wall thickness of the temporary equipment shall be not less than 4.8 mm and the ratio between pipeline and equipment wall thickness should not exceed 1.5;
- existing weld metal and the heat affected zone shall be removed by cut-back over at least 12 mm;
- the full circumference of the prepared weld ends of the pipeline and the pig trap shall be ultrasonically tested to confirm the absence of laminations, over a minimum distance of 25 mm from the bevelled ends;
- welding shall be carried out in accordance with the relevant DEP

Upon completion of pre-commissioning activities, the temporary pig trap facilities shall be removed by cutting out the complete connecting weld and heat affected zone.

5.5. Flexible Hoses

Flexible hoses may be used only for short length connections such as those required to pig traps and instruments. They shall be of the armoured type and be manufactured and tested to recognised standards applicable to permanent hydraulic installations. All hoses and connections shall be of one type only. Prior to each re-use, all hoses and connections shall be visually inspected for damage. Damaged hoses shall be marked accordingly and shall not be used. Hoses shall be protected against external damage and be anchored adequately so that pressure and/or flow variations cannot cause them to jump.

For safety reasons, any temporary or permanent manifolds used in pre-commissioning operations shall be rigidly supported. Vibrating hoses shall not be allowed to transmit their forces into the manifold, otherwise failure due to fatigue may occur leading to potentially serious incidents.

5.6. Electrical Installations

Electrical installations shall comply with the requirements for temporary equipment as given in DEP 33.64.10.10-Gen. The rules governing the use of electrical equipment in hazardous areas shall be complied with. The zone classification shall be as advised by BFPCL.

Equipment shall be earthed for protection against lightning and the accumulation of static electricity. Earth jumper leads should be bonded to equipment before hoses and piping are connected. Hoses and piping should be disconnected and retrieved only with the earth jumper leads connected to equipment.

5.7. Diesel Engines

If diesel engines are used in hazardous areas, their protection shall be considered. No such engines should be used in, or within 15 m of, a Zone 1 area. Their use in a Zone 2 area should be avoided wherever possible. If the use of diesel engines within 15 m of a Zone 1 or 2 area, or within a Zone 2 area, is unavoidable the following shall apply:

- Surface temperature limitation to 250 °C or to the specified temperature class.
- Exhaust flame trap and spark arrestor.
- Non-electric starting system.
- An inlet air shut-down control system to operate in response to high temperature, over speed and gas detection.
- Cooling fan blades should be non-metallic and all fan belts of the antistatic fire-resistant type.
- Any ancillary electrical equipment shall comply with the relevant DEP.

5.7.1. Test Cabin

Test cabins should be air-conditioned or heated, as appropriate, for continuous monitoring and recording, and shall be fully equipped and tested prior to mobilisation to the work sites. Flexible hoses shall not be allowed inside the test cabin.

5.7.2. Instrumentation

All instruments and cabling shall have current certificates of calibration/testing from a recognised certifying authority or third party approved by BFPCL. Calibration/testing certificates shall not be older than 6 months at commencement of work at the work site.

Reading divisions, ranges and accuracies of instruments shall be in accordance with the table below.

INSTRUMENTATION	READING DIVISION	RANGE	ACCURACY
DEWATERING, CONDITION-ING,			
Flowmeter and recorder	m^3/h	(Note 1)	± 2 % (Note 7)
Pressure gauge and recorder	0.5 bar 1 bar	0 bar to 16 bars	± 0.6 % (Note 6)
HIGH VELOCITY WATER, AIR OR			
Flowmeter and recorder for liquids	m^3/h	(Note 1)	± 2 % ± 5 %

Pressure gauge and recorder	0.5 bar 1 bar	0 bar to 16 bars	± 0.6 % (Note 6)
METHANOL OR GLYCOL SWABBING			
Density meter	kg/m^3	(Note 2)	± 1 % (Note 7)
DRYING WITH DRY AIR, DRY GAS			
Dewpoint meter	1.0 °C	+ 40 °C to –50 °C	± 1.0 °C
Ambient air temperature recorder (24 h)	0.5 °C	0 °C to 80 °C	± 1 % (Note 7)
Temperature probes (digital)	0.1 °C	+ 20 °C to –50 °C	± 0.2 °C
Pressure gauge and recorder	0.5 bar	0 bar to 10 bars	± 1 % (Note 7)
Flowmeter and recorder	m^3/h	(Note 3)	± 2 % (Note 7)
VACUUM DRYING			
Barometer	1 mbar	(Note 4)	± 0.8 mbar
Pressure gauge and recorder 1	20 mbar	0 mbar to 1000 mbar	± 1 % (Note 6)
Pressure gauge and recorder 2	1 mbar	0 mbar to 100 mbar	± 1 % (Note 6)
Pressure gauge and recorder 3	0.5 mbar	0 mbar to 50 mbar	± 1 % (Note 6)
Pressure gauge and recorder 4	0.1 mbar	0 mbar to 10 mbar	± 1 % (Note 6)
Flowmeter	m^3/h	(Note 5)	± 2 % (Note 7)
Flowmeter and recorder	m^3/h	(Note 1)	± 2 % (Note 7)
Pressure gauge and recorder	0.5 bar 1 bar	0 bar to 16 bars	± 0.6 % (Note 6)
HIGH VELOCITY WATER, AIR OR			

Flowmeter and recorder for liquids	m^3/h	(Note 1)	± 2 % ± 5 %
Pressure gauge and recorder	0.5 bar 1 bar	0 bar to 16 bar 0 bar to 60 bar	± 0.6 % (Note 6)
METHANOL OR GLYCOL SWABBING			
Density meter	kg/m^3	(Note 2)	± 1 % (Note 7)
DRYING WITH DRY AIR, DRY GAS			
Dewpoint meter	1.0 °C	+ 40 °C to −50 °C	± 1.0 °C
Ambient air temperature recorder (24 h)	0.5 °C	0 °C to 80 °C	± 1 % (Note 7)
Temperature probes (digital)	0.1 °C	+ 20 °C to −50 °C	± 0.2 °C

Table: Reading division, range and accuracy of Instrumentation

NOTES:

i. Depending on flowmeter size (pipeline diameter).
ii. Depending on the drying medium to be used.
iii. Depending on pipeline diameter and/or air-drying unit capacity.
iv. Depending on site altitude.
v. Depending on vacuum plant capacity.
vi. Percentage of measured value.
vii. Percentage of full range.

6. Pigging Requirements

6.1. General

Where required, pigging operations shall comply with the requirements of this Section. Pigs for the pre-commissioning operation shall be selected from the following list:

PIG TYPE	PRECOMMISSIONING APPLICATION	OPTIONAL ATTACHMENTS
Cup	Cleaning Separation Dewatering Batching	Brushes Blades/scrapers Magnets
Disc	Cleaning Separation Dewatering Batching	Brushes Blades/scrapers Magnets
Foam	Cleaning Dewatering Batching Air Drying	Brush/Abrasive Coating Urethane Coating
Sphere	Dewatering Batching	Pig Tracking (with special provision)
Gel	Dewatering Batching	

6.2. Planning Pigging Operations

Pipeline elevation, density of fluid in the pipeline, pressure loss across the pig or pig train and back-pressure at the receiving end shall be considered when calculating the required inlet flows and pressures for the driving medium.

The pressure loss across a pig is mainly dependent upon the pig type and line diameter but is also influenced by the flowing medium and the pipe's internal roughness. As an approximate guide for the pipelines, a pressure loss of 0.5 bar should be allowed, increasing up to 3 bar for smaller than NPS 16 diameters; where the available driving pressure could become a critical factor, a more exact value should be obtained from the pig manufacturer.

Control of the pig travelling speed shall be assisted by maintaining a back-pressure of 1 bar (ga) for liquids and 3 bar (ga) for other media above the ambient pressure at the receiving end. This pressure shall be increased where necessary to counteract acceleration of the pig in downhill sections. In liquid-filled pipelines, the pressure of the receiving end shall always be equal to or greater than the static pressure that can occur behind the pig during pigging operation.

Single pigs or pig trains shall be separated by distances that provide sufficient time for receivers to be unloaded without reducing the travelling speed of any following pigs or pig trains. Pig trains shall be split if they contain more pigs than the receiver can hold before being unloaded.

The distance between pigs in a train shall be at least 2 % of the pipeline length, with a minimum distance of 150 metres when separated by liquids and 300 metres when separated by other media.

Prior to any pigging operations the Contractor shall verify that:

- all valves have been correctly positioned and are functional. All main line valves shall be in the fully open position; and
- discharge and/or storage facilities have been connected and are in good working order.

Pigs should be removed from the pig receiver immediately upon arrival, and then inspected.

6.3. Pig Train Monitoring

During all pigging operations the location of the pig trains should be predicted by calculations and measurement of the volume of the driving medium. Launching and arrival of pigs at pig traps should be monitored by pig signallers fitted to the pig traps or by measuring signals from pig location devices fitted to the last pig of the pig train.

When glycol or other chemicals are part of the pig train or used as the driving medium, the location of pig trains should be monitored during the pigging operations. Location monitoring should be performed by monitoring the signals from pig location devices or by predictions based on the volumes of the discharged fluids measured in the discharge pipe work.

Pig location devices should be of the magnetic, radio or ultrasonic types; for liquid filled offshore pipelines the acoustic Pinger type may also be considered.

6.4. Dislodging Stuck Pigs

Attempts to dislodge pigs shall be aborted before the applied pressure exceeds either:

a. MAOP of the pipeline/pre-commissioning equipment; or
b. the predicted pigging operation pressure by:

    - 50 % for lines filled completely with liquids; or
    - 5 bar (ga) for all other media.

The Contractor shall advise BFPCL when higher pressures are required to dislodge pigs and shall proceed in accordance with the contingency procedures only after approval by BFPCL.

6.5. Pigging Records

All pigging operations shall be recorded in a pig data register with the following data as a minimum:

- identification number and date of the pig run;
- number and type of pig;

- volumes and pressures of the driving media;
- time of launching and receiving;
- condition of pigs before launching and upon arrival; and
- volumes and natures of material and substances arriving in front of the pig at the receiving end.

7. Dewatering

7.1. General

Water shall be removed or inhibited to prevent hydrate formation and pipeline corrosion based on gas composition and pipeline conditions.

When specified by BFPCL in the scope of work, Contractor shall dewater the pipeline using the specified medium.

Final water cleaning may be combined with the first dewatering run. The Contractor shall perform final water cleaning before or in combination with the first dewatering run if required.

Only high sealing disc pigs with at least two guiding and four sealing discs should be used unless otherwise specified in the scope of work.

Sleeping pigs shall be recovered by the Contractor. These pigs shall be used for dewatering only if specified in the scope of work.

The Contractor shall drain remaining free water from valve body cavities and dead ends of piping and if applicable blow through with compressed air. Where drying is specified, failure to perform this activity will prolong drying operations and could also result in localised icing up if vacuum drying is applied. The method and timing of draining shall be specified in the dewatering procedure of the pre-commissioning plan.

Dewatering of the pipeline by gravity alone is not permitted, as this can result in a broken water column which can seriously affect the pressure needed to propel pigs due to the accumulation of multiple static heads.

7.2. Dewatering Runs

All pipelines should be dewatered by a pig train consisting of at least two high sealing pigs. One pig train run shall be required as a minimum when the driving medium is:

- nitrogen, glycol or methanol;
- nitrogen in combination with glycol swabbing.

Two pig train runs shall be required as a minimum when the driving medium is:

- air;
- air in combination with glycol swabbing.

Where drying is to follow dewatering, it is important to measure both the outflow of water and inflow of air during dewatering in order to perform a mass/volume balance. This will enable the quantity of residual water in the pipeline to be estimated. This estimation may be used as the basis for drying time predictions.

NOTE: Methanol shall not be used in dewatering operations where air drying is to follow, as this can lead to potentially explosive methanol/air mixtures. Glycol may be used with tri-ethylene glycol being preferred due to its lower toxicity level.

7.2.1. First Dewatering Run

The first dewatering run shall be carried out as follows:

- the travelling speed of the pig train should be constant and within the range of 0.5 to 1.0 m/s;
- the internal pipeline pressure shall be kept at least 1 bar (ga) above the ambient pressure at any point along the pipeline by control of back-pressure at the discharge end;
- when dewatering with air or nitrogen only, pigs shall be separated;
- when dewatering with methanol or glycol only, pigs shall be separated by a volume of the driving medium to be determined.

Fresh water swabbing shall be performed if salt water was introduced into the pipeline for hydrostatic pressure testing or pre-commissioning and final air cleaning or drying is required. This is to prevent salt deposits forming, especially during evaporation. The requirements for fresh water swabbing are:

- fresh water, of about 4 % of the pipeline volume, shall be driven through the pipeline in batches contained by pigs;
- the volume of fresh water between two pigs shall be determined;
- the travelling speed and internal pipeline pressure shall be as specified above.

If the fresh water batches are driven by air, swabbing shall be repeated until the salt content of a received batch is less than 200 mg per kg of water. If the driving medium is glycol or methanol, or is followed by glycol swabbing, swabbing should be stopped once the specified volume of fresh water has been pigged through the pipeline.

7.2.2. Second dewatering run

A pig train with one high sealing disc pig, followed by at least by two foam pigs with a density of 80 kg/m3, shall be driven by air through the pipeline as follows:

- the dewpoint of the air for pigging shall be at least 5 °C below the minimum ambient temperature along the pipeline measured at 1 bar (ga);
- the travelling speed of the pigs should be between 0.5 m/s and 1.0 m/s, but shall not exceed 1.5 m/s;
- pigs shall be separated in accordance with standard requirements.

Pigging with foam or brush foam pigs shall continue until they arrive without free water in front.

Foam pigs should not be used in long non-internally coated pipelines due to the quantity of foam dust remaining in the pipeline, as this has a detrimental effect on the subsequent drying operation.

7.3. Dewatering Records

- The Contractor shall measure and record the following:
    - details of test water and other media received at receiving end;
    - salt content of the last fresh water batch at the receiving end, if applicable;
    - for each dewatering run the quantities, pressure and flowrate records of the driving medium;
    - for the second run the dewpoints of the air.

8. Final Cleaning

The Contractor shall adopt the method for final cleaning as specified by BFPCL in the scope of work.

Final water cleaning shall be carried out before or in combination with the first dewatering run.

Low velocity air cleaning shall be carried out in combination with dry air pigging. High velocity air cleaning should be carried out only after the pipeline has been fully dried in accordance with specified requirements.

BFPCL shall specify the cleanliness criteria to be achieved consistent with the selected cleaning method and the intended pipeline service. Cleanliness criteria shall be as follows:

- o When the cleaning method involves the use of pigs, the amount of debris received at the end of each pig run shall be assessed on a volume and mass basis. The debris received shall show a reducing trend, in both volume and particle size. Pigging shall continue until it is demonstrated that further pigging will not result in a significant reduction in the debris received.
- o Regular samples shall be taken from the effluent. The solids content shall be measured and the results recorded. Cleaning shall be considered complete when the quantity of debris in the samples directly ahead of the last pig is less than one percent by volume.
- o The pipeline system shall be considered clean when the water/air coming out of the pipeline does not leave any debris on a filter with size 50 μm (300 mesh).

8.1. Low Velocity Water Cleaning

Low velocity water cleaning shall be carried out by pigging the pipeline with pig trains consisting of a series of brush cleaning pigs followed by a high sealing disc pig, as follows:

- clean and filtered water shall be used as driving medium;
- high sealing disc pigs should be run with open bypass ports of approximately 8 % of the pipeline bore;
- the pig travelling speed should be within the range 0.5 m/s to 1.0 m/s.

Pigging shall continue until the specified cleanliness criteria have been achieved. Final cleaning of carbon steel pipelines shall be concluded by a run with a magnetic cleaning pig.

8.2. High Velocity Water Cleaning

High velocity water cleaning shall be carried out by continuously pumping water at high velocities through the pipeline, as follows:

- clean and filtered water shall be pumped continuously through the pipeline at a minimum velocity of 3 m/s;
- the minimum volume to be pumped through the line shall be three times the volume of the pipeline.

Cleaning shall be considered complete when the specified cleanliness criteria have been achieved whilst replacing the line content at the specified velocity.

8.3. Low Velocity Air Cleaning

Low velocity air cleaning shall be carried out in combination with pigging. Pig trains consisting of a series of brush cleaning pigs or brush coated foam pigs followed by a high sealing disc pig shall be driven through the pipeline using dry air, as follows:

- high sealing disc pigs should be run with open bypass ports of approximately 5 % of the pipeline bore;
- the pig travelling speeds should be within 0.5 m/s to 1.0 m/s;
- the dewpoint of the air shall be as for drying.

Pigging shall continue until the specified cleanliness criteria have been achieved. Final cleaning of carbon steel pipelines shall be concluded by a run with a magnetic cleaning pig.

8.4. High Velocity Air Cleaning

High velocity air cleaning shall be carried out after dry air or vacuum drying. Dry air shall be purged continuously through the pipeline as follows:

- the dewpoint of the dry air shall be as specified for the dry pipeline;
- a minimum air velocity of 10 m/s shall be maintained in the pipeline at the inlet;
- the quantity of air to be used for purging shall be sufficient to replace the volume of the pipeline three (3) times at the purging pressure.

Cleaning shall be complete when the specified cleanliness criteria have been achieved whilst the line content is replaced at the specified velocity. When using this method, special attention shall be given to knocking out any entrained debris/dust before the air is exhausted to the atmosphere.

8.5. Final Cleaning Records

The Contractor shall measure and record the following:

- total volume, pressure and flowrate of cleaning medium for each cleaning step;
- quantities and nature of materials received at the end of the pipeline;
- details of media received at the receiving end;
- for air cleaning the dewpoints of the air from both ends of the pipeline.

9. Drying

9.1. Introduction

The Contractor shall adopt the drying method and achieve the dryness as specified in the scope of work. The general requirements for drying are:

- dry air, nitrogen or vacuum drying shall commence immediately after completion of the second dewatering run;
- the second dewatering run shall be repeated if drying of the pipeline has not commenced within 48 h of completion of the dewatering;
- valve body cavities and dead ends of piping shall be dried in compliance with the method and timing defined in the pre-commissioning plan;
- the sizing of the drying equipment and calculations of the time required for drying shall be based on a film thickness of the residual water of not less than 0.1 mm for internally uncoated pipes and not less than 0.05 mm for internally coated pipes. Lower values may be assumed if the Contractor can demonstrate, to BFPCL's satisfaction, their
- validity from metering during dewatering or from previous experience.

9.2. Dry Air and Nitrogen Drying

9.2.1. General

The drying medium to be used shall be as specified in the scope of work. Dry air or nitrogen drying shall be executed consecutively in the following phases:

- pigging;
- purging for drying; and
- purging for acceptance testing.

Air introduced into the pipeline during dry air drying shall have a dewpoint of at least 15 °C below the final dewpoint of the pipeline as specified in the scope of work.

Nitrogen used during drying shall have a minimum dewpoint of 50 °C at atmospheric pressure.

This method is often protracted, requires considerable equipment and space, is noisy, is heavy on fuel for compression and is unsuitable for pipelines with “dead-legs” or subsea terminations.

9.2.2. Pigging

The pipeline shall be pigged with high sealing disc pigs driven by dry air or nitrogen in combination with water absorbing foam pigs having a large water absorption capacity (approximately 80 % of their body mass), high abrasion resistance and a density between 30 to 50 kg/m3 as follows:

- the travelling speed of the foam pigs should not exceed 1.2 m/s;

- a back-pressure of at least 0.5 bar (ga) shall be maintained at the receiving end; and
- pigs in a pig train should be separated by at least 300 linear pipeline metres.

Pigging shall continue until the dewpoint of the drying medium at the receiving end remains below the dewpoint specified in the scope of work and does not fluctuate by more than 3°C whilst replacing the content of the line by a pig.

9.2.3. Purging for drying

After pigging, the pipeline should be purged with the drying medium with a minimum velocity of 3 m/s in the pipeline at the discharge end. Purging shall continue until the dewpoint at the discharge end remains below the specified dewpoint whilst replacing twice the content of the pipeline at purging pressure.

9.2.4. Purging for acceptance testing

The difficulty in defining the acceptance criterion is that the dewpoint sampling at each end of the pipeline does not necessarily represent the actual dewpoint condition prevailing within the whole pipeline. This is because the dry air (or nitrogen) entering the pipeline performs extensive drying at the start of the pipeline and then becomes saturated. As the pressure falls off towards the end of the pipeline, the air (or nitrogen) is again able to absorb moisture. Thus the situation can arise where the beginning and the end of a pipeline are dry but the middle may still be wet, or at a higher dewpoint than the ends. It can be checked that the acceptance criterion has been met by means of the following procedure.

Upon completion of purging, the pipeline shall be blocked-in for a period of at least 12 h and at a pressure of 0.5 bar (ga) above the ambient pressure at all points along the pipeline. After this period the pipeline content shall be replaced at the lowest possible pressure and the dewpoint continuously measured at the discharge end.

Drying is complete when the dewpoint during acceptance purging remains below the final dewpoint specified for the pipeline whilst replacing the line content. Purging for drying shall recommence and the acceptance test shall be repeated until this requirement has been met.

Upon completion of the drying, the pipeline shall be blocked in at a pressure of 0.5 bar (ga) above the ambient pressure at any point along the pipeline.

9.3. Vacuum Drying

9.3.1. General

Water vapour shall be evacuated from the pipeline by vacuum units alone or in combination with dry air or dry nitrogen vacuum purging as specified in the scope of work. The final dewpoint of the dry pipeline shall be 20 °C, which is equivalent to a pressure of 1 mbar (abs), unless otherwise specified in the scope of work.

Vacuum drying should consist of the following stages unless otherwise approved by BFPCL:

- pre-drying checks;
- one or more leak tests;
- pump-down;
- evaporation/evacuation, including vacuum purging;
- soak test/acceptance test;
- filling phase.

The size of the vacuum units should be sufficient to reduce the pressure in the pipeline to the vaporisation pressure during pump-down within typically 12 h to 36 h of commencement of the pump-down operation, depending on the length and diameter of the pipeline. Vacuum units having excessive capacity would draw down the pressure too rapidly, which could cause localised ice formation.

The Contractor shall prepare a theoretical pressure/time graph for each of the drying phases, which shall be included in the pre-commissioning plan.

9.3.2. Pre-drying checks

Before commencement of vacuum drying the Contractor shall verify that:

- the pipeline has been isolated from other pipelines and pipework;
- valves are designed for vacuum drying and have been placed in to the half open position;
- valve body bleeder parts are vacuum tight;
- temporary connections, pig trap valves and pig trap end closure seals are able to withstand the prevailing vacuum pressure. If this is not the case, Contractor shall provide adequate seals for the vacuum drying operation and replace these seals by the permanent seals once the vacuum drying operation has been completed.

9.3.3. Leak tests

The pipework connecting the vacuum unit with the pipeline, including pig trap and vacuum unit(s), shall be isolated from the pipeline and the pressure in the isolated pipework lowered to slightly above the theoretical evaporation pressure.

NOTE: For dry pipework a typical leak test pressure of 8 mbar (abs) to 10 mbar (abs) should be applied.

The vacuum unit shall be switched off and the isolated pipework checked for leaks by visual/audio inspection. Leaks shall be cured by flange tightening, taping flange circumferences, taping grease nipples and applying plasticine or equivalent to valve stems.

The pipeline shall then be opened to the vacuum unitand the pressure in the entire system reduced to a pressure of 50 mbar (abs) to 100 mbar (abs) for the final leak test. The pressure shall be maintained at this level and all other piping, such as at inline block valve stations and the pig trap system at the other end of the pipeline if vacuum drying is carried out from one end only, shall be checked for leaks. Leaks shall be cured as stated above.

After all leaks have been cured, where possible, the vacuum unit shall be turned off and isolated from the pipeline and the pressure in the pipeline and the associated pipework monitored for at least 1 h. Pressure increases shall be recorded and plotted on a pressure/time chart. From the measured pressure increase, the total leak rate shall calculate. Curing of leaks shall be continued until the calculated total leak rate is less than

10 % of evacuation capacity of the vacuum units at the initial leak test pressure. The final in-leak rate shall be recorded for use when analysing the final soak test results.

9.3.4. Pump-down

The pressure in the pipeline shall then be reduced at a steady rate to the level predicted for the evaporation of the residual water. A significantly shorter pump down time than that theoretically predicted could indicate freezing and shall be evaluated immediately. The pressure shall be kept at this level, and pig traps and piping inspected for vacuum tightness and any leaks cured.

9.3.5. Evaporation/evacuation

As the pressure in the pipeline approaches the saturated vapour pressure at the pipeline's ambient temperature, the rate of vapour evolution will increase, resulting in a reduction in the rate of pressure decrease. The vaporisation pressure shall be maintained and water vapour evacuated by pumping until all residual water has evaporated. Once all the free water has evaporated from the pipeline, the rate of pressure decrease will increase.

Ice formation in the pipeline and associated fittings shall be avoided by control of the evacuation rate through the vacuum units. A vaporisation pressure plateau at a level markedly lower than expected or erratic pressure fluctuations during plateau are indications of ice formation.

NOTE: The temperature of the water vapour drawn from the pipeline should remain above +3 °C at atmospheric pressure.

Vaporisation and evacuation by pumping shall continue until the vapour pressure has reached the level that is equivalent to the dewpoint specified for the dry pipeline.

This pressure shall be maintained for at least 3 h to confirm that a stable balanced vacuum pressure is established throughout the pipeline. Evacuation shall then be stopped and a soak test carried out.

Vacuum purging with dry air or nitrogen at pressures in the range of 3 mbar (abs) to 10 mbar (abs) may be applied in addition to evacuation by pumping to reduce the time needed for conventional evaporation and water vapour evacuation. The rates and pressures are dependent on the performance curves of the vacuum equipment, as the aim is to increase the pressure in the pipeline to an efficient volume transfer level. If applied, purging and evacuation shall continue until the dewpoint at the vacuum unit is constantly below the dewpoint for a dry pipeline as specified in the scope of work while replacing at least twice the contents of the pipeline. Purging shall then be stopped, and the pressure reduced to 3 mbar (abs) and maintained at this level for at least 3 h to achieve stable conditions in the pipeline. A soak test shall then be performed.

9.3.6. Soak Test/Acceptance Test

The pipeline shall be blocked in and isolated from all equipment other than that required for pressure monitoring for a period of at least 24 h, at a pressure of:

- 1 mbar (abs) if vapour is being evacuated by pumping only;
- 3 mbar (abs) if vapour is being evacuated by pumping and purging;

or as specified in the scope of work.

Pressure monitoring shall be carried out by means of pressure gauges and recorders with range 0 mbar to 10 mbar, a reading division of 0.1 mbar and an accuracy of ± 1 % of the measured value.

Initially the pressure will rise as the higher pressure in the centre of the pipeline (or at the opposite end if a single vacuum plant is in operation) balances with that nearest to the vacuum plant. After this initial stabilisation, which should occur well below the evaporation plateau, it shall be demonstrated that any further observed pressure increases are attributable to the leaks measured during the pipeline leak test (8.3.3). If this is not the case, the observed pressure increases must be due to further flashing-off of moisture vapour, indicating that additional drying is required.

9.3.7. Filling phase

After the soak test has been successfully completed, the Contractor shall fill the pipeline with either dry air or dry nitrogen as specified in the scope of work, to:

- 50 mbar (abs), for nitrogen when gas filling is to immediately follow drying.
- 500 mbar (ga) above the ambient pressure at any point along the pipeline, for nitrogen (if there will be a delay between drying and gas filling) or for dry air.

Warning signs, in English and the local or working languages, such as "PIPELINE UNDER VACUUM" or "PIPELINE FILLED WITH DRY AIR" shall be placed at all affected pipework, including block valve stations and pig trap systems.

9.3.8. Dry Air Pigging In Combination With Vacuum Drying

Dry air pigging followed by vacuum drying shall be carried out after completion of the second dewatering phase, as follows:

- the dewpoint of the dry air shall be 40 oC unless otherwise specified in the scope of work.
- dry air pigging shall be performed and shall be continued until the dewpoint of the air received is 15 oC at atmospheric pressure and remains at this level whilst the line content is replaced by a dry air driven pig;
- after completion of dry air pigging, vacuum drying in combination with dry air vacuum purging in compliance with (8.3) shall follow.

10. CONDITIONING AND PRESERVATION

10.1. PRESERVATION AFTER DRYING

The Contractor shall increase the pressure in the pipeline with the same medium as specified in this document. The requirements for preservation are as follows:

- the final pipeline pressure to be achieved at the end of the filling operation shall be 0.5 bar (g) above the ambient pressure at any point along the pipeline, plus a margin allowing for the maximum possible ambient temperature fluctuation during the post pre-commissioning period;
- the dewpoint, pressure and temperature of the medium introduced into the pipeline shall be measured and recorded constantly at the inlet of the pipeline throughout the filling operation;
- Warning signs, in English and the local or working languages, such as "PIPELINE FILLED WITH NITROGEN" or "PIPELINE FILLED WITH DRY AIR" shall be provided and placed at block valve stations and pig trap systems.

10.2. CONDITIONING AND PRESERVATION RECORDS

The Contractor shall provide product certificates of the media introduced into the pipeline plus the following parameters as a minimum:

- - quantities introduced into and received from the pipeline;
- - batch volume and pigging records, if applicable;
- - dewpoint, temperature, pressure and flowrates of the medium being used.

11. Pre-commissioning Checklist

Description	Requirement / value
Commencement of pre-commissioning Immediately after hydrostatic pressure testing? or immediately prior to commissioning? Specify the date:	Yes [ ] No [ ] Yes [ ] No [ ] ....
Sequence of pre-commissioning To be in accordance with this DEP? If no, state the required sequence:	Yes [ ] No [ ] ....
Pipeline data State the relevant pipeline data:	
Outside diameter:	.... mm
Nominal wall thickness:	.... mm
Minimum bend radius:	.... mm
Minimum bore:	.... mm
Length:	.........................m
Internal volume:	......................m^3
MAOP:	...............bar (ga)
Maximum / Minimum Design pressure:	....../........bar (ga)
Maximum / Minimum Design temperature:	......../ ....°C
Linepipe material:	.... Yes [ ] No
Flowcoating:	.... Yes [ ] No
Pig traps:	.... Yes [ ] No
Block valve stations: If yes, state number:	.... Yes [ ] No
Non-return valves and/or special fittings. If yes, state type and number	.... Yes [ ] No
Hydrotest water data	
Potable Water used for line-fill water	.... Yes [ ] No

Seawater used for line-fill water	.... Yes [ ] No
Are chemicals or a dye added to the line-fill water? If yes, state details:	Yes [ ] No [ ] ....
Hazardous area classification	
Do hazardous areas exist at pre-commissioning work sites? If yes, state the area classification:	Yes [ ] No [ ] Zone.................. Zone.................. Zone..................
Timing of the submission of the pre-commissioning plan	
To be in accordance with DEP? If no, specify the submission time of the plan in advance of the planned date of the first activity at the work site:	Yes [ ] No [ ] .... days
Pre-commissioning records and reports	
Are more than three copies of the final report required? If Yes, state the number of copies: Is the submission date in accordance with DEP? If no, state working days:	Yes [ ] No [ ] .... copies Yes [ ] No [ ] .... days
Gas detection devices & breathing apparatus Are gas detection devices required? If yes, state the number: Is breathing apparatus required?	Yes [ ] No [ ] .... Yes [ ] No [ ]
Maximum allowable noise level As specified in DEP? If no, specify the maximum allowable noise level:	Yes [ ] No [ ] .... dB (A)
Filters, pumps and compressors Are filters of <50 micrometres required? If yes, specify size required	Yes [ ] No [ ] ……………micrometres

Temporary pig traps Are temporary pig traps required? If yes, state number of pigs to be stored and length:	Yes [ ] No [ ] ....pigs ....m
Welding of temporary equipment As specified in this DEP? If no, state welding specification to be used.	Yes [ ] No [ ] ....
Driving medium of dewatering operations Is compressed air the driving medium? If no, specify the medium:	Yes [ ] No [ ] ....
Dewatering pigs As specified in DEP? If no, state type:	Yes [ ] No [ ] ....
Dewatering with sleeping pigs Are sleeping pigs in the pipeline? If yes, state type and the number of pigs	Yes [ ] No [ ] ....
Purity of the fluid Is contamination of the fluid with chlorides acceptable?	Yes [ ] No [ ]
Duration of the post dewatering period To be in accordance with this DEP? If no, specify the duration:	Yes [ ] No [ ] .... days
Final cleaning Is final cleaning required? If yes, state selected cleaning process and method: If water cleaning, state velocity: If air cleaning, state velocity:	Yes [ ] No [ ] .... High [ ] Low [ ] High [ ] Low [ ]

Drying Is drying required? If yes, state the drying process: Dry air or nitrogen drying? Vacuum drying? Combination of drying methods? Methanol or glycol swabbing?	Yes [ ] No [ ] Yes [ ] No [ ] Yes [ ] No [ ] Yes [ ] No [ ] Yes [ ] No [ ]
Dry air or nitrogen drying If yes, state the medium: Is the medium dry air? Is the medium nitrogen? Is the dewpoint in accordance with DEP? If no, specify dewpoint required:	 Yes [ ] No [ ] Yes [ ] No [ ] Yes [ ] No [ ] ....°C
Vacuum drying	Yes [ ] No [ ]
General Is the evaporation/evacuation phase in combination with vacuum purging? If yes, state the medium: Dry air? Dry nitrogen? Are the dewpointin accordance with DEP? If no, specify the vacuum pressure required:	Yes [ ] No [ ] Yes [ ] No [ ] Yes [ ] No [ ] ....mbar (abs)
Filling phase Is filling as specified in DEP? If no, specify the pressure:	Yes [ ] No [ ] ....mbar (abs)
Dry air pigging in combination with vacuum drying If yes, is the dry air dewpoint in accordance with DEP? If no, specify:	Yes [ ] No [ ] ....°C
Glycol swabbing If selected, state: Is the acceptance criterion in accordance with DEP? If no, specify water content of the drying medium:	 Yes [ ] No [ ] ....mass %

Conditioning by Inhibition Is inhibition required? If yes, state: Film thickness: Type of Inhibitor: Volume: Type of carrier medium: Volume:	Yes [ ] No [ ]..................... mm ..../....................... m3 ..../........................ m3

11.1.1. Summary of Required Approvals By BFPCL

Activity / Procedure requiring approval
Pre-commissioning plan Pre-commissioning plan
Pre-commissioning records and reports Final draft of the "Final Pre-commissioning Report"
Health, Safety and Environment Work site location and arrangement of equipment Water supply and disposal locations and facilities Emergency plan
Pig selection Sealing gel specification and pigging procedure
Dislodging stuck pigs Removal of stuck pigs from the pipeline